

## # Document Control

- System Name: Application
- Version: 1.0
- Author: Data Engineering Team
- Date: 2024-06-12
- Status: Draft

## # Executive Summary

- The Application facilitates an automated ETL pipeline involving source CSV data ingestion, transformation, cleaning, aggregation, and SCD Type 2 history management, implemented using Delta Live Tables.

## # Business Objectives

- Automate data ingestion from CSV files to Delta tables
- Ensure data quality through null and duplicate removal
- Aggregate customer spend and maintain accurate, auditable transaction history
- Enable change data capture via SCD Type 2 implementation

## # In-Scope

- Source CSV data reading and delta table loading
- Schema enforcement for customer and order tables
- Data transformation and cleaning logic
- Creation of ordersummary and customeraggregatespend tables
- Aggregation logic on ordersummary data
- SCD Type 2 implementation for ordersummary
- Process deployment using Delta Live Tables

## # Out-of-Scope

- Any manual data manipulation
- External system integrations beyond defined delta tables
- Real-time or streaming ingestion
- Data visualization or reporting dashboards

## # Stakeholders

- Data Engineering Team
- Business Analysts
- Compliance Team

## # Business Requirements

### ### BR-ETL-001 — Load Source Customer Data

- Description: Read customer CSV data and load to delta table /Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata
- Business Value: Automated, structured customer data ingestion
- Priority: High

### ### BR-ETL-002 — Load Source Order Data

- Description: Read order CSV data and load to delta table /Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata
- Business Value: Automated, structured order data ingestion
- Priority: High

### ### BR-ETL-003 — Customer Table Schema Enforcement

- Description: Enforce schema with fields CustId, Name, EmailId, Region for customer table
- Business Value: Consistency and data quality assurance
- Priority: High

### ### BR-ETL-004 — Order Table Schema Enforcement

- Description: Enforce schema with fields OrderId, ItemName, PricePerUnit, Qty, Date, CustId for order table
- Business Value: Consistency and data quality assurance
- Priority: High

### ### BR-ETL-005 — Calculate TotalAmount Field

- Description: Add column TotalAmount to order table ( $\text{PricePerUnit} * \text{Qty}$ )
- Business Value: Simplifies downstream aggregation and analytics
- Priority: Medium

### ### BR-ETL-006 — Data Quality: Null and Duplicate Removal

- Description: Remove nulls and duplicates from customer and order tables
- Business Value: Improves analytical accuracy and reliability
- Priority: High

### ### BR-ETL-007 — Create Ordersummary Table

- Description: Create ordersummary table in catalog=gen\_ai\_poc\_databrickscoe, schema=sdlc\_wizard if not exists
- Business Value: Centralized storage for enriched order data
- Priority: High

#### ### BR-ETL-008 — SCD Type 2 on Ordersummary

- Description: Apply SCD Type 2 methodology to ordersummary based on updates in customer table
- Business Value: Ensures historical accuracy and traceability of data changes
- Priority: High

#### ### BR-ETL-009 — Create CustomerAggregateSpend Table

- Description: Create customeraggregatespend table with columns Name, TotalAmount, Date if not exists
- Business Value: Enables spend analysis and reporting
- Priority: Medium

#### ### BR-ETL-010 — Aggregate Customer Spend

- Description: Aggregate TotalAmount grouped by Name and Date from ordersummary and load into customeraggregatespend
- Business Value: Facilitates spend tracking for business insights
- Priority: Medium

#### ### BR-ETL-011 — Delta Live Tables Implementation

- Description: Implement the entire processing steps using Delta Live Tables
- Business Value: Simplifies workflow orchestration and monitoring
- Priority: High

#### # Assumptions & Constraints

- Source data is available and accessible via defined CSV file paths
- Only the specified schemas will be enforced
- Delta Lake infrastructure is operational

#### # Success Metrics

- Successful, automated runs from CSV to target tables
- Zero null/duplicate records post-processing
- Accurate aggregation results in customeraggregatespend
- SCD Type 2 audit trails present

# Risks & Mitigations

- Source data format changes — Mitigation: Schema enforcement and validation
- Data quality issues from source — Mitigation: Robust null and duplicate handling logic
- Infrastructure/permission errors — Mitigation: Pre-execution checks and monitoring

# Approval

- Business Owner: Pending
- Data Engineering Lead: Pending
- Compliance Review: Pending
- Date: Pending

# Appendix

Appendix A: Source Definitions

| Source Name  | Type | Location                                                   | Format | Notes                              |
|--------------|------|------------------------------------------------------------|--------|------------------------------------|
| customerdata | CSV  | /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata | CSV    | Raw customer data, schema-enforced |
| orderdata    | CSV  | /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata    | CSV    | Raw order data, schema-enforced    |

Appendix B: Target Definitions

| Table Name                                                  | Columns                                                                 |
|-------------------------------------------------------------|-------------------------------------------------------------------------|
| gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary           | OrderId, ItemName, PricePerUnit, Qty, Date, CustId, TotalAmount, SCD Co |
| gen_ai_poc_databrickscoe.sdlc_wizard.customeraggregatespend | Name, TotalAmount, Date                                                 |

Appendix C: Transformations

| Transformation                               | Input                   | Output                 | Logic                              |
|----------------------------------------------|-------------------------|------------------------|------------------------------------|
| Enforce customer schema                      | customerdata            | customerdata           | CustId, Name, EmailId, Region      |
| Enforce order schema & calculate TotalAmount | orderdata               | orderdata              | Add TotalAmount = PricePerUnit * Q |
| Remove nulls/duplicates                      | customerdata, orderdata | cleaned tables         | Null/duplication logic             |
| Aggregate spend                              | ordersummary            | customeraggregatespend | Group by Name, Date; sum TotalA    |
| Create SCD Type 2                            | ordersummary            | ordersummary           | Manage change history on custome   |

Appendix D: Business Rules

| Rule ID    | Area      | Description                                           | Source             |
|------------|-----------|-------------------------------------------------------|--------------------|
| BR-ETL-001 | Ingestion | All customer data must be loaded nightly from CSV     | Data Engineering   |
| BR-ETL-004 | Schema    | Order table must match predefined schema              | Compliance         |
| BR-ETL-008 | SCD       | Ordersummary must capture all changes with versioning | Business/Data Team |

## Appendix E: Process Flow

| Step | Description                          | Dependency              |
|------|--------------------------------------|-------------------------|
| 1    | Ingest customer/order CSVs           | Source availability     |
| 2    | Apply schema and clean data          | Successful ingestion    |
| 3    | Calculate aggregates & create tables | Cleaned data            |
| 4    | Apply SCD Type 2 updates             | Ordersummary generation |
| 5    | Store final outputs                  | Table creation & SCD    |

Figure 1: End-to-End Data Pipeline Flow

Figure 2: SCD Type 2 Processing Logic

